Strict monotonicity of a transformed gamma median

An analytic proof with an exact finite certificate

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Abstract

Let $m(k)$ denote the median of the gamma distribution with shape $k > 0$ and scale one. We prove that $A(k) = 2^{1/k}m(k) - k$ has negative derivative for every $k > 0$. This resolves the particular monotonicity assertion left conjectural by Lyon in 2023. The proof combines elementary analytic estimates on $(0, 1/10]$ and $[7, \infty)$ with 1,052 certified secant inequalities on $[1/10, 7]$. The secant argument uses the previously proved strict convexity of the gamma median. Its numerical inputs are 1,483 rational median brackets, independently verified by a supplied standard-library program using integer arithmetic and explicit series remainders. The proof depends on established gamma-median theorems; it is computer assisted, internally checked, and neither formally verified in a proof assistant nor externally peer reviewed.

1 Statement and mathematical dependencies

For $k > 0$, define

$$P(k, x) = \frac{1}{\Gamma(k)} \int_0^x t^{k-1} e^{-t} \, \mathrm{d}t, \qquad P(k, m(k)) = \frac{1}{2}. \tag{1}$$

The density is positive on $(0, \infty)$, so the median exists and is unique. Throughout this paper the scale is exactly one. Put

$$c = \log 2, \qquad q(k) = e^{-c/k}, \qquad b(k) = \frac{m(k)}{q(k)}, \qquad A(k) = b(k) - k. \tag{2}$$

Theorem 1. *For every real $k > 0$,*

$$A'(k) < 0.$$

Consequently $A(k_1) > A(k_2)$ whenever $0 < k_1 < k_2$.

Lyon [1] proved sharp constant bounds on this exact transform and explicitly left its monotonicity conjectural. Our proof uses the following established properties:

$$0 < m'(k) < 1, \qquad m''(k) > 0, \qquad k - \frac{1}{3} < m(k) < k \quad (k > 0). \tag{3}$$

The continuous Chen–Rubin result of Berg and Pedersen [2, Theorem 3.4] gives the derivative inequality; their strict-convexity theorem [3, Theorem 1.1] gives the second. The lower median bound is also stated in [2, p. 63], with attribution to Chen and Rubin. Alternatively it follows from $m' < 1$ and the limit $m(k) - k \to -1/3$ proved in [2, Proposition 4.2]. The upper bound follows from $m' < 1$ and $m(0+) = 0$. For the latter limit, Markov's inequality gives $m(k) \le 2k$. These are cited mathematical inputs, not claims of new results here. The 2004 arXiv version of [2] numbers its derivative theorem 3.3; we use the journal numbering.

On compact subsets of $k > 0$, the defining integral and its parameter derivatives are dominated by integrable functions with factors $t^{\alpha-1}(1 + |\log t|^j)$ near zero. The implicit

function theorem, since $P_x(k, m(k)) > 0$, therefore justifies the derivatives used here. In particular,

$$A'(k) = e^{c/k}\left(m'(k) - \frac{c\, m(k)}{k^2}\right) - 1. \tag{4}$$

No derivative of a formal asymptotic remainder is used.

2 The small-parameter interval

Introduce, for independent variables $k, m > 0$,

$$T(k,m) = k\int_0^1 u^{k-1}e^{-mu}\,\mathrm{d}u, \quad J(k,m) = -\frac{\log T(k,m)}{k}, \quad h(k) = \frac{\log\Gamma(1+k)}{k}. \tag{5}$$

Substitution $t = m(k)u$ in the median equation yields $m(k)^k T(k, m(k)) = \Gamma(1+k)/2$ and hence

$$\log b(k) = h(k) + J(k, m(k)). \tag{6}$$

Subscripts on J below denote partial derivatives.

Lemma 2. *For $k, m > 0$,*

$$J_k(k,m) \le 0, \qquad 0 < J_m(k,m) \le \frac{e^m}{k+1}, \qquad h'(k) < \frac{33}{40}.$$

Proof. Integration by parts gives

$$T(k,m) = e^{-m} + m\int_0^1 u^k e^{-mu}\,\mathrm{d}u.$$

For fixed m, this is the Laplace transform in k of a probability measure: an atom of mass e^{-m} at zero, together with the image under $u \mapsto -\log u$ of the measure $me^{-mu}\,\mathrm{d}u$. Its total mass is $e^{-m} + 1 - e^{-m} = 1$. Thus $f(k) = \log T(k,m)$ is convex and $f(0) = 0$. Convexity gives $0 \ge f(k) - kf'(k)$, whence $J_k \le 0$. For completeness, the convexity follows from Cauchy–Schwarz applied to the Laplace integral, or by expressing f'' as the variance of the integration variable under exponential tilting.

Differentiation under the integral sign gives

$$J_m(k,m) = \frac{\int_0^1 u^k e^{-mu}\,\mathrm{d}u}{T(k,m)} \le \frac{e^m}{k+1},$$

because $T(k,m) \ge e^{-m}$. Writing $\psi = (\log\Gamma)'$ and using the trigamma series,

$$h'(k) = \int_0^1 t\psi'(1+kt)\,\mathrm{d}t \le \frac{\psi'(1)}{2} = \frac{1}{2}\sum_{n=1}^{\infty}\frac{1}{n^2} < \frac{33}{40}.$$

The last inequality follows without a decimal approximation from $\sum_{n=1}^{10} n^{-2} + 1/10 < 33/20$, using the integral test. All differentiations above follow from domination on compact parameter intervals. □

Proposition 3. *For $0 < k \le 1/10$,*

$$A'(k) \le -\frac{35273}{4087200} < 0.$$

Proof. Log-convexity of Γ and $\Gamma(1) = \Gamma(2) = 1$ imply $h(k) \le 0$ for $0 < k \le 1$. If U has density ku^{k-1} on $(0,1)$, Jensen's inequality gives

$$T(k,m) = \mathbb{E}(e^{-mU}) \ge e^{-m\mathbb{E}U} = e^{-mk/(k+1)}.$$

Apply this with $m = m(k)$. Equations (6) and (3), and $e^x < 1/(1-x)$ for $0 < x < 1$, imply

$$b(k) \le e^{m(k)/(k+1)} < e^{k/(k+1)} \le e^{1/11} < \frac{11}{10}.$$

Since $q(k) \le 2^{-10} = 1/1024$, we obtain

$$m(k) < \frac{11}{10240}, \quad J_m < \frac{1}{1-m(k)} \le \frac{10240}{10229}, \quad 1 - m(k)J_m \ge \frac{10218}{10229} > 0. \tag{7}$$

We use the elementary enclosure

$$\frac{69}{100} < c < \frac{7}{10}. \tag{8}$$

Indeed the first two terms of $\log 2 = 2\sum_{n\ge 0}((2n+1)3^{2n+1})^{-1}$ give $\log 2 > 56/81 > 69/100$; and $e^{7/10} > 1 + 7/10 + 49/200 + 343/6000 > 2$ proves the other side. Since

$$\left(\frac{q(k)}{k^2}\right)' = \frac{q(k)(c-2k)}{k^4} > 0 \qquad (0 < k \le 1/10),$$

we have

$$\frac{c\,m(k)}{k^2} \le \frac{7}{10}\frac{11}{10}\frac{100}{1024} = \frac{77}{1024}. \tag{9}$$

Now $m' = mb'/b + cm/k^2$. Differentiate (6) to get

$$\frac{b'}{b}(1 - mJ_m) = h' + J_k + \frac{cm}{k^2}J_m.$$

Use Lemma 2 and (7)–(9). The denominator is positive, and the numerator has a positive upper bound, so

$$b' \le \frac{11}{10}\frac{33/40 + 770/10229}{10218/10229} = \frac{4051927}{4087200} < 1.$$

Subtracting one proves the proposition. □

3 The large-parameter interval

Proposition 4. *For every $k \ge 7$, $A'(k) < 0$.*

Proof. Set $r = c/k$. By (8) and $k \ge 7$,

$$\frac{1}{2} - \frac{1}{3c} - \frac{r}{6} \ge \frac{1}{2} - \frac{1}{3c} - \frac{c}{42} > \frac{1}{2} - \frac{100}{207} - \frac{1}{60} = \frac{1}{4140} > 0.$$

Taylor's theorem with its positive fourth-order remainder gives

$$e^{-r} > 1 - r + \frac{r^2}{2} - \frac{r^3}{6} > 1 - r + \frac{r^2}{3c}.$$

Using $m' < 1$ and $m > k - 1/3$ in (4),

$$A'(k) < e^r\left(1 - r + \frac{r^2}{3c}\right) - 1 < 0.$$

In fact the argument gives the explicit upper bound $A'(k) < -e^{c/k}c^2/(4140k^2)$. □

4 A finite certificate for the middle interval

The following bridge is what turns certified values into a proof on intervals. Checking only pointwise derivative values would not be sufficient.

Lemma 5. *Let $0 < a < b < d$, and suppose nonnegative rational numbers L_a, L_b, U_d satisfy $L_a \le m(a)$, $L_b \le m(b)$, and $m(d) \le U_d$. Define*

$$D = \frac{U_d - L_b}{d - b}, \qquad B = D - \frac{cL_a}{b^2}, \qquad V = e^{c/a} \max(B, 0).$$

Then $A'(k) \le V - 1$ for every $k \in [a, b]$.

Proof. Strict convexity from (3) implies

$$m'(k) \le m'(b) < \frac{m(d) - m(b)}{d - b} \le D.$$

Since m increases, $m(k) \ge L_a$, and $k^2 \le b^2$. Thus the parenthesized quantity in (4) is at most B. If $B \le 0$, this gives $A'(k) \le -1 \le V - 1$. If $B > 0$, use $e^{c/k} \le e^{c/a}$ to obtain the assertion. □

Proposition 6. *For every $k \in [1/10, 7]$,*

$$A'(k) < -\frac{27}{10^6} < 0.$$

Computer-assisted proof. The accompanying certificate contains a sorted partition into 1,052 closed intervals $[a, b]$, and for each a rational $d > b$. The endpoints join exactly, starting at $1/10$ and ending at 7. It also contains positive rational lower and upper bounds L_s, U_s at every one of the 1,483 distinct nodes s used as a, b, d. Every d is at most $701/100$. The verifier establishes, with strict rational inequalities,

$$P(s, L_s) < \tfrac{1}{2} < P(s, U_s), \tag{10}$$

$$e^{c/a} \max\left(\frac{U_d - L_b}{d - b} - \frac{cL_a}{b^2}, 0\right) < \frac{999973}{10^6} \quad \text{for every row.} \tag{11}$$

Monotonicity of the CDF in its second argument validates the median brackets. Lemma 5, applied to each closed interval, then proves the proposition including all connecting endpoints. Section 5 supplies the full arithmetic method and error bounds for these finite checks. □

Certificate property	Verified value or bound
Covered interval	$[1/10, 7]$
Number of closed intervals	$1,052$
Number of rational median brackets	$1,483$
Largest auxiliary node	$701/100$
Uniform upper bound for every V	$999973/10^6$
Small-parameter derivative bound	$-35273/4087200$
Large-parameter derivative bound	$-e^{c/k}c^2/(4140k^2)$

Table 1: The proof covers the whole positive half-line. The finite table is certified interval by interval, not inferred by interpolation.

Proof of Theorem 1. Propositions 3, 6, and 4 cover $(0, \infty)$, with overlap at $1/10$ and 7. Each gives a strictly negative derivative. The mean value theorem on any compact interval $[k_1, k_2] \subset (0, \infty)$ gives the asserted strict decrease. □

5 Exact arithmetic and explicit remainders

The generation and verification computations are separate. Arb, through python-flint, was used to locate suitable median brackets and split intervals until a sufficient bound was found. Its output is only candidate data. The independent verifier uses Python's arbitrary-size integers, rational inputs, and outward-rounded dyadic intervals. It does not use Arb, floating-point arithmetic, or a library implementation of a special function in its certification path.

5.1 Interval operations and elementary functions

An interval is represented by integers (ℓ, u) denoting $[\ell/2^{192}, u/2^{192}]$. Each elementary rational operation is rounded outward by integer floor and ceiling. Multiplication uses the minimum and maximum of the four endpoint products; division rejects any denominator interval containing zero. Every rational input is similarly enclosed. Consequently an induction on the arithmetic operations preserves containment.

For logarithms, range reduction writes $x = 2^j y$ with $1 \le y \le 2$. Set $t = (y-1)/(y+1)$, so $0 \le t \le 1/3$. After N terms,

$$\log y = 2\sum_{n=0}^{N-1} \frac{t^{2n+1}}{2n+1} + R_N, \quad 0 \le R_N \le \frac{2t^{2N+1}}{(2N+1)(1-t^2)}. \tag{12}$$

The same formula encloses $\log 2$, and $\log x = j\log 2 + \log y$. General input intervals are evaluated at their endpoints using monotonicity.

For exponentials, divide the argument by a power of two until its absolute value is at most $1/8$. If $|r| \le 1/8$, then

$$\left| e^r - \sum_{n=0}^{32} \frac{r^n}{n!} \right| \le \frac{2|r|^{33}}{33!}. \tag{13}$$

Here Taylor's remainder uses $e^{|r|} < 2$. Repeated interval squaring reverses the range reduction. Finally π is enclosed by Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, with the usual alternating series remainder bounded by the first omitted term.

5.2 The gamma function

For positive real z, the Stirling expansion with $N-1$ terms is

$$\begin{aligned}\log\Gamma(z) = {} & (z-\tfrac{1}{2})\log z - z + \tfrac{1}{2}\log(2\pi) \\ & + \sum_{j=1}^{N-1} \frac{B_{2j}}{2j(2j-1)z^{2j-1}} + R_N(z),\end{aligned} \tag{14}$$

where $R_N(z)$ has the sign of B_{2N} and magnitude at most $|B_{2N}|/[2N(2N-1)z^{2N-1}]$. This is the standard positive-real signed remainder [5, §5.11(ii)]. One derivation uses Binet's formula

$$\log\Gamma(z) = (z-\tfrac{1}{2})\log z - z + \tfrac{1}{2}\log(2\pi) + 2\int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t}-1}\,\mathrm{d}t.$$

The finite geometric expansion of the derivative of $\arctan u$, integrated from zero to u, has a remainder of alternating sign bounded by its next monomial for every $u \ge 0$. Integrating it against the positive kernel above proves the stated sign and bound and yields the Bernoulli coefficients in (14).

For $\log\Gamma(k+1)$ the verifier takes $z = k+33 > 33$, uses $N = 25$ (terms through B_{48}, with the B_{50} remainder), and then applies the exact recurrence

$$\log\Gamma(k+1) = \log\Gamma(k+33) - \sum_{j=0}^{31} \log(k+1+j).$$

The Bernoulli numbers are computed as exact rationals.

5.3 Gamma CDF enclosures

For $k, x > 0$, integration by parts or termwise differentiation gives the convergent positive series

$$P(k,x) = e^{k\log x - x - \log\Gamma(k+1)} \sum_{n=0}^{\infty} t_n, \quad t_0 = 1, \quad t_{n+1} = \frac{x}{k+n+1} t_n. \tag{15}$$

To verify the identity directly, differentiate $e^{-x}x^k \sum_{n\geq 0} t_n$ with respect to x: the terms cancel to give $kx^{k-1}e^{-x}$, and the expression tends to zero as $x \downarrow 0$. The series converges locally uniformly. After term t_n, if $r = x/(k+n+1) < 1$, the remaining ratios are at most r, so

$$0 < \sum_{j=n+1}^{\infty} t_j \leq \frac{t_{n+1}}{1-r}. \tag{16}$$

Equations (12)–(16), with outward interval operations, prove both signs in (10). They also enclose every expression in (11). If any sign is unresolved, the program fails; it does not accept an approximate equality or silently assume a bound.

5.4 Certificate structure and reproduction

The supplement contains two CSV tables. Each row of `certificates/median_brackets.csv` consists of (s, L_s, U_s), written as exact rational strings. Each row of `certificates/intervals.csv` consists of (a, b, d); the third column is named `c` in the file, solely as a column label, and is distinct from the mathematical constant $c = \log 2$.

The acceptance algorithm is:

```
For every distinct node s:
    Require 0 < L_s < U_s.
    Prove P(s,L_s) < 1/2 < P(s,U_s).
Read the interval rows in their supplied order.
Require first a=1/10, last b=7, and adjacent b=a.
For every (a,b,d):
    Require 0<a<b<d and all three nodes in the table.
    D = (U_d-L_b)/(d-b).
    Prove exp(log(2)/a)*max(D-log(2)*L_a/b^2,0)
          < 999973/1000000.
Accept only if every requirement passes.
```

Run the independent checker from the extracted project directory:

```
python src/rational_verify.py
```

Only the Python standard library is needed. The complete code, the tables, and a machine-readable verification report accompany the paper. Optional regeneration requires python-flint:

```
python src/generate_certificate.py
```

Regenerating the candidate data is not necessary to verify the proof. With pdfLaTeX available, rebuild the paper using:

```
python src/build_paper.py
```

The released checker was also run on a clean copy to test that these relative paths and supplied files suffice.

6 Examples, scope, and attribution

The exactly known value $m(1) = \log 2$ gives

$$A(1) = 2\log 2 - 1 \approx 0.3862943611.$$

Known endpoint limits [2, 4, 1] are

$$\lim_{k\downarrow 0} A(k) = e^{-\gamma} \approx 0.5614594836, \qquad \lim_{k\to\infty} A(k) = \log 2 - \frac{1}{3} \approx 0.3598138472.$$

Together with Theorem 1, these recover the strict constant bounds proved earlier by Lyon. The endpoint limits serve as examples and context; they are not substitutes for the analytic tail estimates in the proof.

The exact scope is the median, probability level $1/2$, at scale one and every positive real shape parameter. We do not claim the corresponding statement for arbitrary quantiles or other transforms. The literature audit, accessed on 28 October 2026, included Lyon's paper and later related work, Berg–Pedersen, and Pedersen's transformed-quantile paper. No earlier proof of this exact monotonicity assertion was found in that bounded search. This statement records the search scope, not a guarantee of worldwide novelty.

The new contribution of this investigation is the combination of the explicit two-tail inequalities with the exact finite secant certificate. Classical median convexity, the continuous Chen–Rubin theorem, and gamma-function identities retain their original attribution. The derivation, code, and manuscript were produced by GPT-6 Astra, an AI system, and reviewed through independent internal derivations and a separate arithmetic verifier. No external review or proof-assistant certification is asserted.

References

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